Cover Letter

To: The Editor-in-Chief
Journal: *Remote Sensing* (MDPI)
Date: October 16, 2026

Dear Editor,

We are pleased to submit our manuscript entitled "The Split Is Part of the Measurement: Spatial-Interleaving Evaluation Bias in Change Detection, Quantified in Bi-Temporal Mangrove Mapping from Sentinel-2" for consideration for publication in *Remote Sensing*.

This paper documents a spatial-evaluation failure mode in bi-temporal change detection: a spatially interleaved random split produces a 29.6-percentage-point higher change $F1$ than a buffer-separated test assignment, positive in all ten partitions, while the corresponding ROC-AUC difference is only $+1.9$ pp. The audit quantifies this in a Sentinel-2 mangrove change-detection setting (0.8–2.6% change prevalence), decomposes the gap into a small prevalence-arithmetic component ($+2.7$ pp) and a dominant recall shift associated with the joint spatial configuration of training and test samples, and states the identification boundary of what the design can and cannot attribute.

We believe this work is well suited to *Remote Sensing* because it addresses a methodological concern that affects a large share of change-detection studies: the evaluation protocol itself moves the reported score. The paper provides a quantified, reproducible demonstration of the effect size, a decomposition that separates prevalence arithmetic from the spatial-configuration component, and a ten-item reporting checklist for practitioners.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have read and approved the submitted version. The full code base and results are deposited at `https://github.com/hao-feier/split-audit`.

Thank you for your consideration.

Sincerely,

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